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Epistaxis and Nasal trauma

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learning objective

- Understanding to define epistaxis and nasal fracture.
- Able to categorize types of epistaxis and nasal fracture
- Learning how to approach the patient with nasal fracture and epistaxis
- Enhancing knowledge regarding how to manage patients in emergency with nasal fracture and epistaxis

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Content:

- Anatomy
- Definition & Epidemiology
- Blood supply
- Classification
- Etiology
- Management
- Case study

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Anatomy

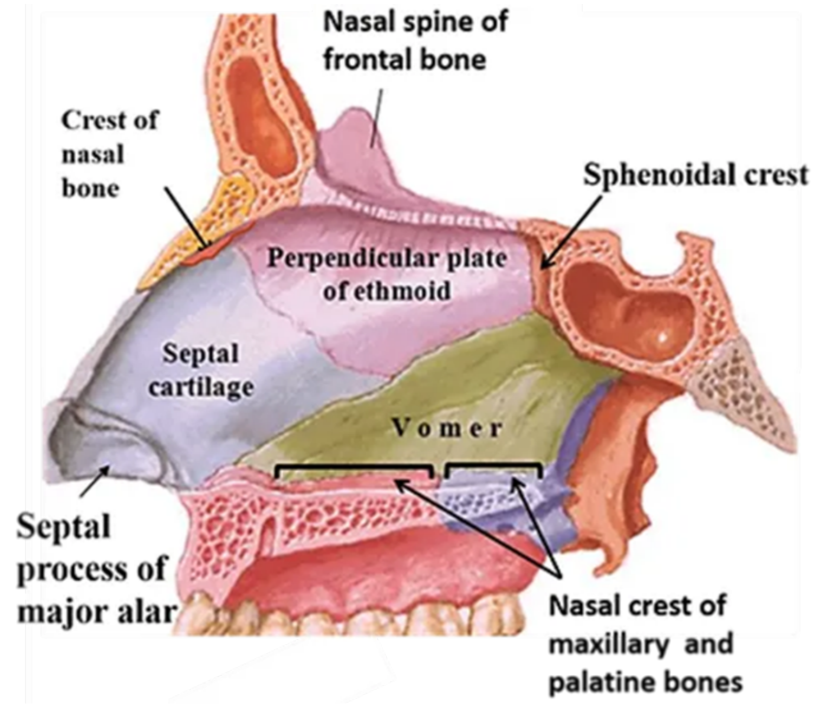
- **<u>External nose:-</u>** it composes of
- <u>bone</u>
- <u>Cartilages</u>
- <u>skin,soft tissues and muscles</u>
- **<u>Internal nose or nasal cavity:-</u>** nasal cavity divided by a <u>septum</u> into right and left sides.

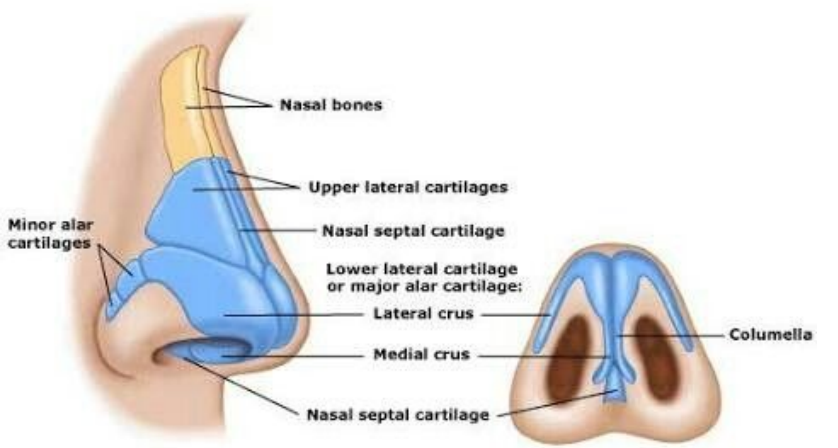
The nasal septum is composed of three parts:

-<u>membranous septum</u>

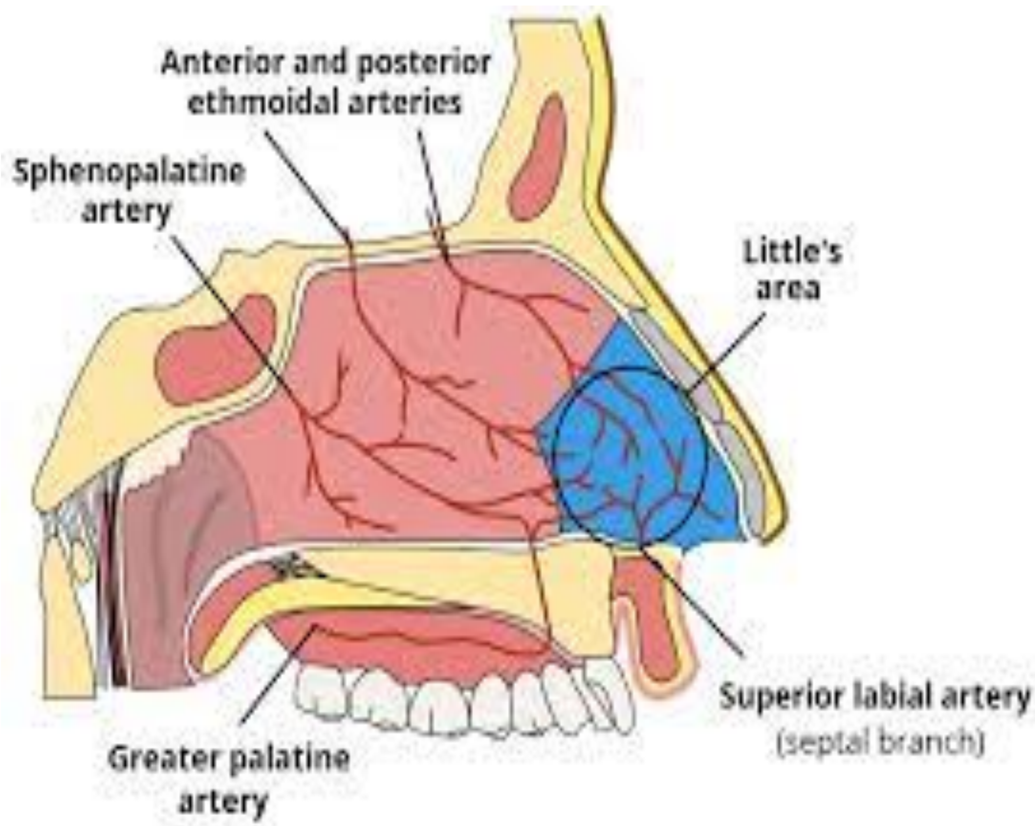
-<u>quadriangular cartilage</u> or <u>septal cartilage</u>

-<u>bony septum</u>





Anatomy-Arterial supply



Definition & Epidemiology

- A nose bleeding is the loss of blood from the tissue that lines inside of your nose. Common ENT emergency.
- Lifelong incidence of epistaxis in general population is about 60%
- Fewer than 10% seek medical attention
- Peaks in young children (2 – 10 y) and older individuals (50 – 80 y)

Classification

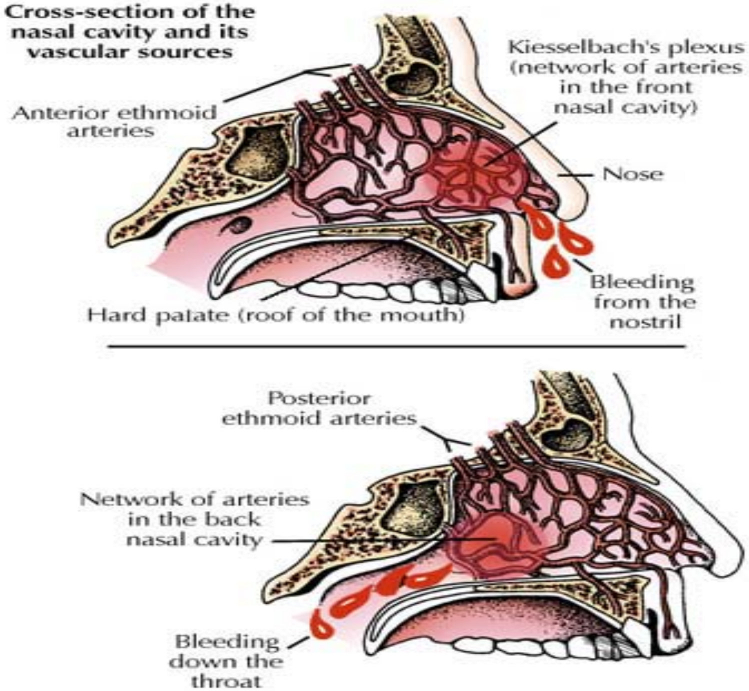
- **According to the site:-**

-Anterior epistaxis: any bleeding anterior to pyriform aperture.

-Posterior epistaxis: bleeding posterior to pyriform aperture.

Classification

- Anterior
  - 90% of all cases of epistaxis
  - Kiesselbach's plexus
  - Younger population
  - Typically less severe
- Posterior
  - Woodruff's plexus
  - Older population
  - Associated with bleeding from both nostrils



Classification

Childhood and Adulthood onset:-

There is a pronounced bimodal distribution in the age of onset of epistaxis.

The condition is common in childhood, becomes less common in early adult life and then peaks in the sixth decade.

Thus, while it can occur at any time in life, this variation with age is sufficiently pronounced to classify epistaxis as childhood (less than 16 years) or adult (greater than 16 years).

Classification

- Primary or secondary:-

-Between 70% and 80% of all cases of epistaxis are idiopathic, spontaneous bleeds without any proven precipitant or causal factor.

-This type of bleeding can be classified as primary epistaxis.

-A small proportion of cases are due to a clear and definite cause such as trauma, surgery, or anti-coagulant overdose and can be classified as secondary epistaxis.

Etiology

A-Most are idiopathic (primary epistaxis)

B-Local causes(secondary epistaxis)

1-Trauma

- Nose picking/blowing, sneezing, fractures

2-Foreign bodies

3-Iatrogenic

- FESS, rhinoplasty, nasal cannula

4-Inflammation/infection(viral rhinitis, nasal diphtheria, acute rhino sinusitis)

5-Tumors

carcinoma/angiofibroma

6-Hereditary telangiectasia

7-deviated nasal septum(DNS)

C-Systemic causes

- Hepatic cirrhosis, Uremia, Wegener’s granulomatosis.

D-Coagulopathy

- Hemophilia, von Willebrand’s, Thrombocytopenia, Hematological malignancy

E-Drugs

- NSAIDs, aspirin, warfarin, heparin

F- weather

- Dry cold polluted weather

Management

- Control significant bleeding or hemodynamic instability before obtaining a lengthy history
- Steps:
  - First aid and resuscitation
  - Assess blood loss
  - Localize bleeding
  - Control bleeding

First Aid & Resuscitation

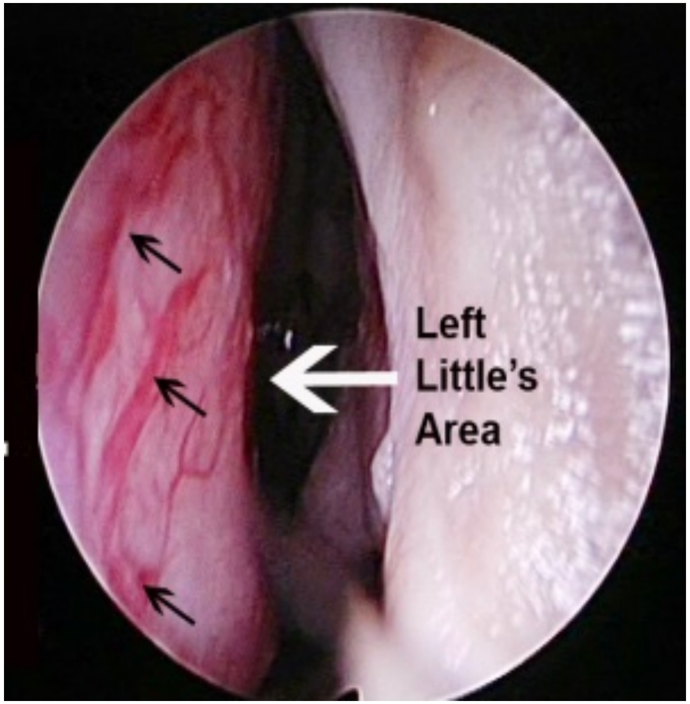
- Vital signs and signs of shock
- Patient with significant. hemorrhage should receive an IV line and crystalloid infusion.
- Cross-match for packed RBC.
- Baselineinvestigations

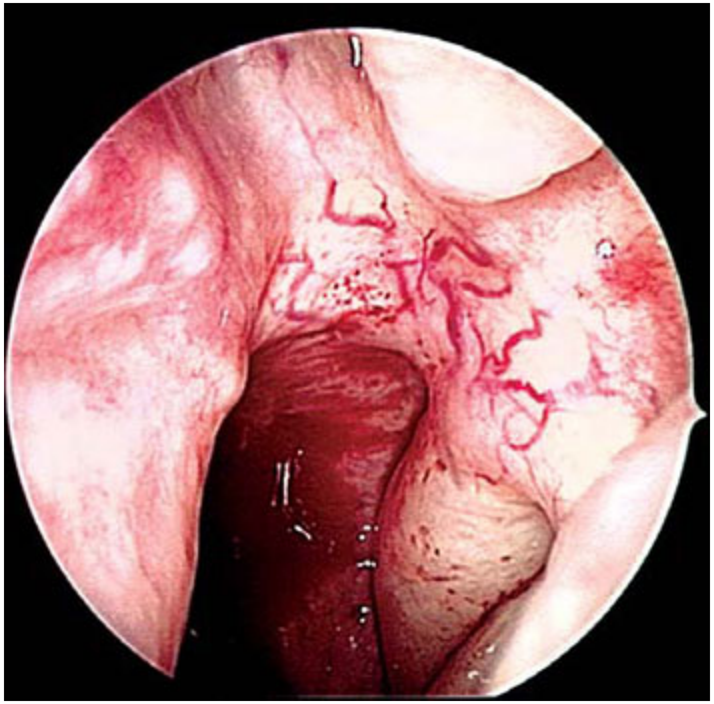
(CBC,Coagulation profile,Renal fuction test)

Localization of Bleeding

- Examination of the nose by speculum or endoscopy to locate the site of bleeding

- Pledgets soaked with anesthetic-vasoconstrictor solution are inserted into the nasal cavity to anesthetize and shrink nasal mucosa
- Allow them to remain for 10 – 15 minutes


Left
Little's
Area



Control of Bleeding

-Direct therapy: when bleeding point identified.

- *Chemical cauterization* with silver nitrate stick or trichloro-acetic acid
  - Rolled over mucosa until a grey eschar forms
  - Only one side should be cauterized to prevent septal necrosis or perforation

- *Thermal cauterization* with an electrocautery device for more aggressive bleeding under LA or GA(bipolar or unipolar cautery??)


Grafco®
Silver Nitrate Applicator—6"
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Silver Nitrate..............75%
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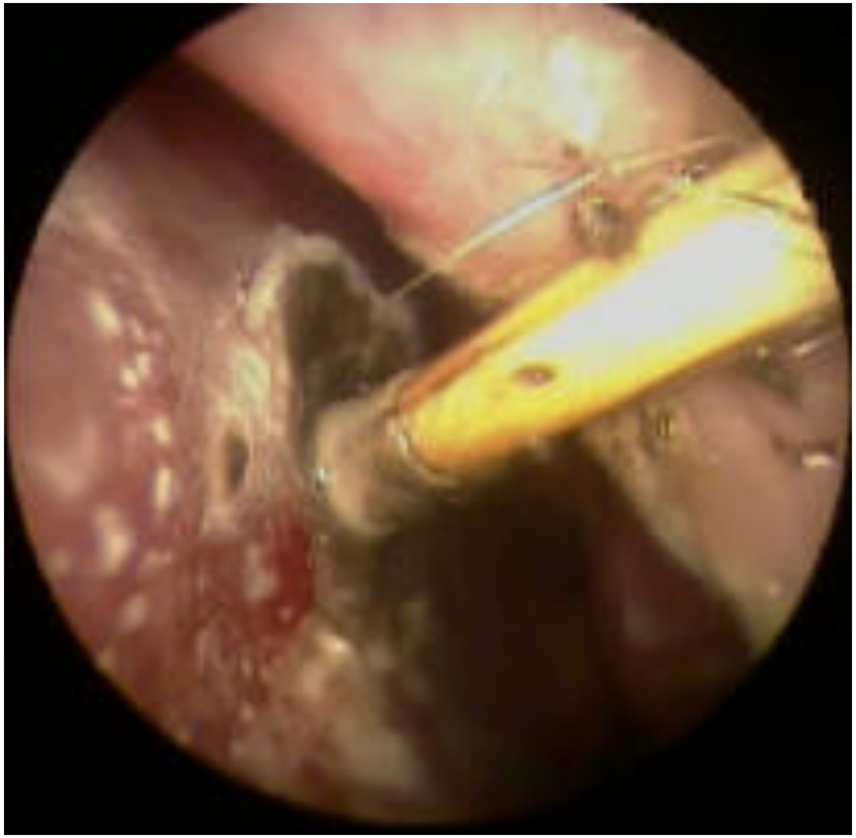
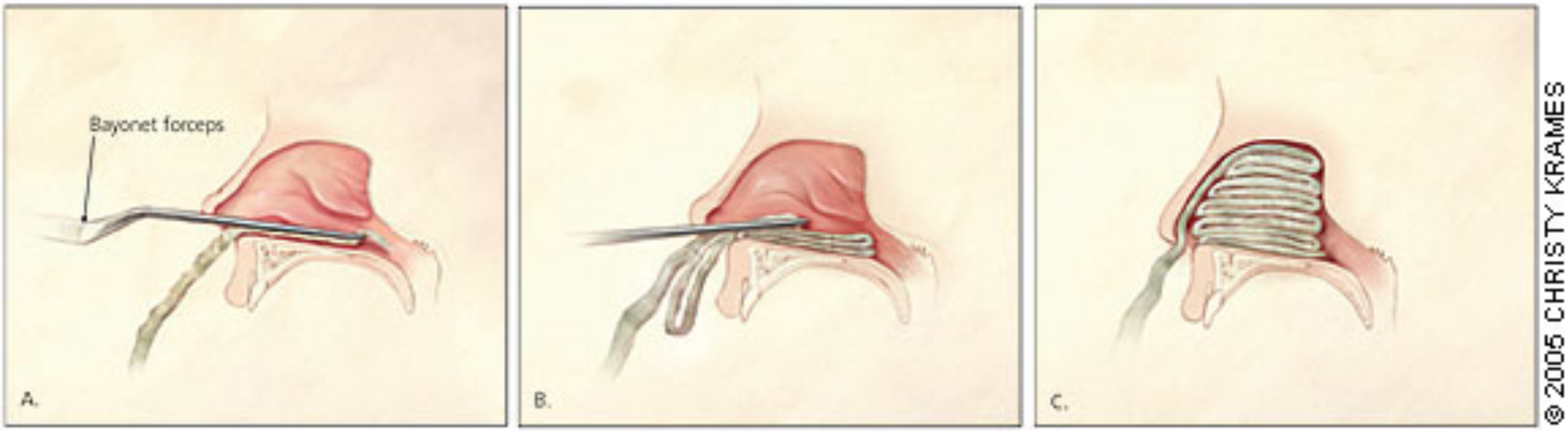


-Anterior Nasal Packing

- Traditional petrolatum gauze filled with antibiotic ointment
- Success rate 85%, let it for up to 48 hours under antibiotic coverage


Bayonet forceps
A.
B.
C.
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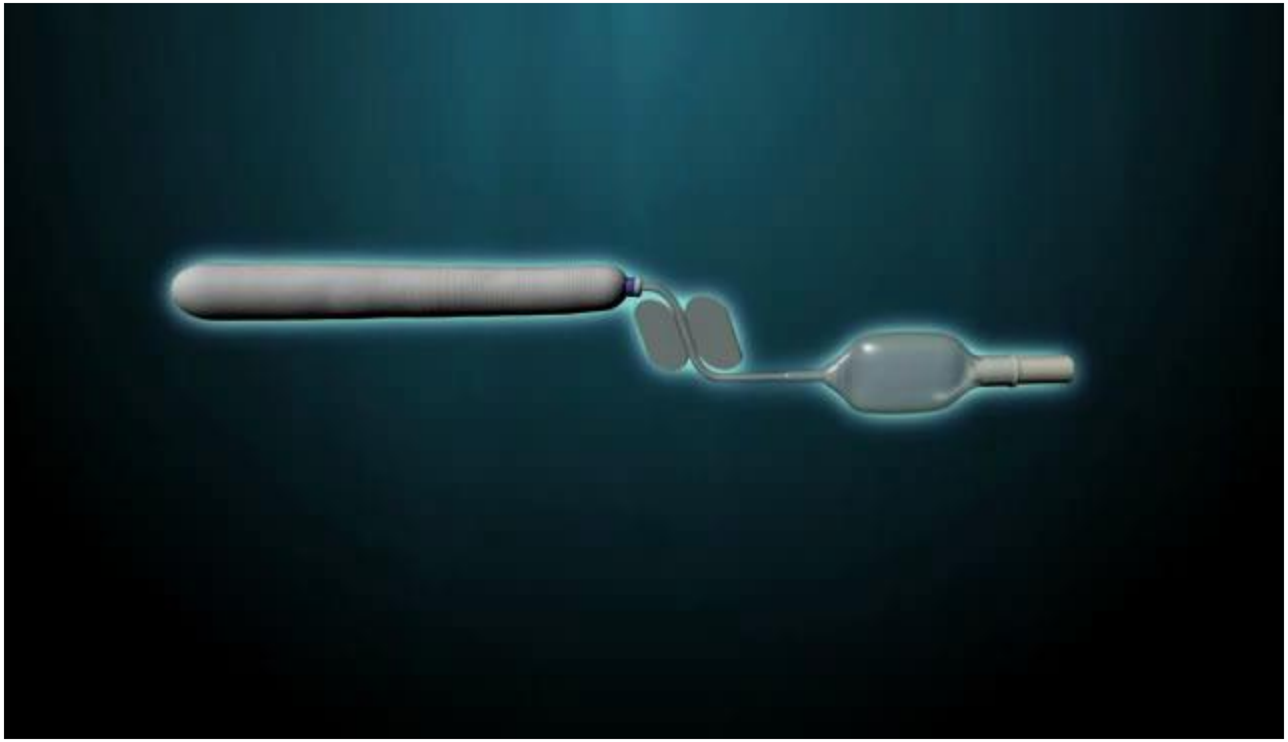
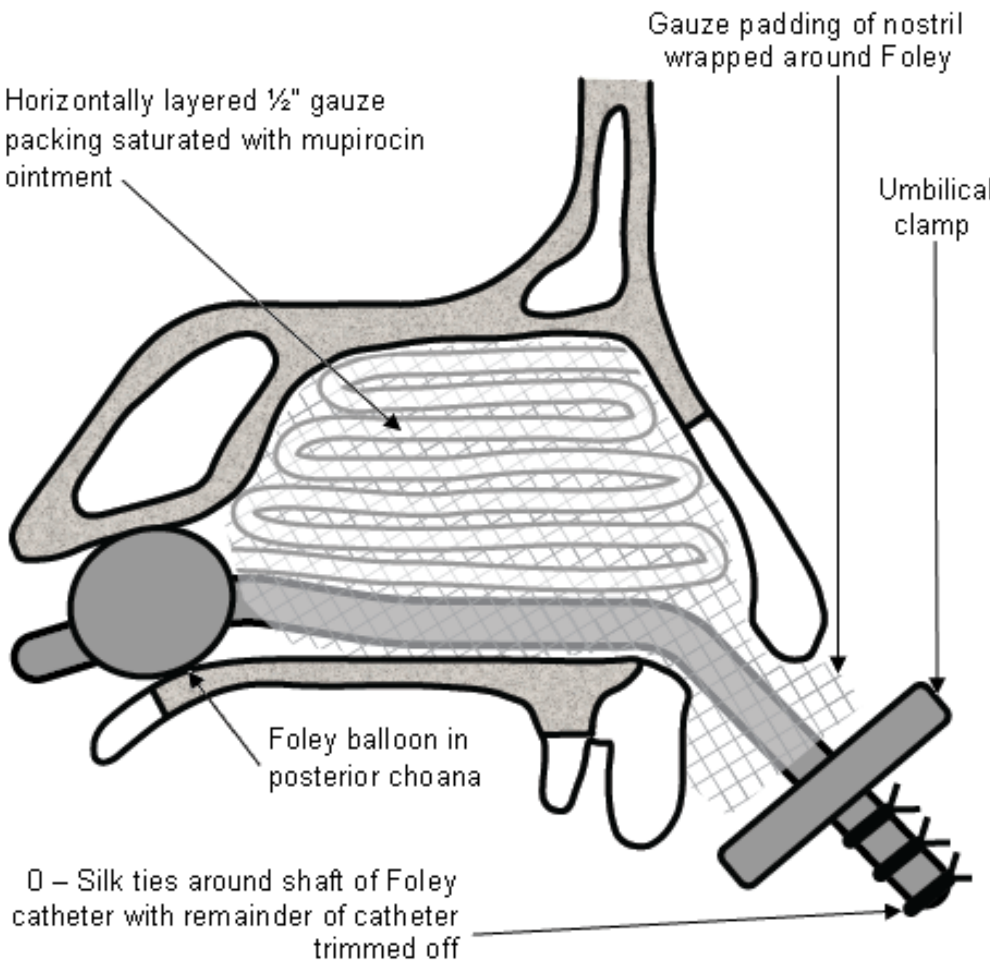
- Rapid Rhino anterior balloon tampon



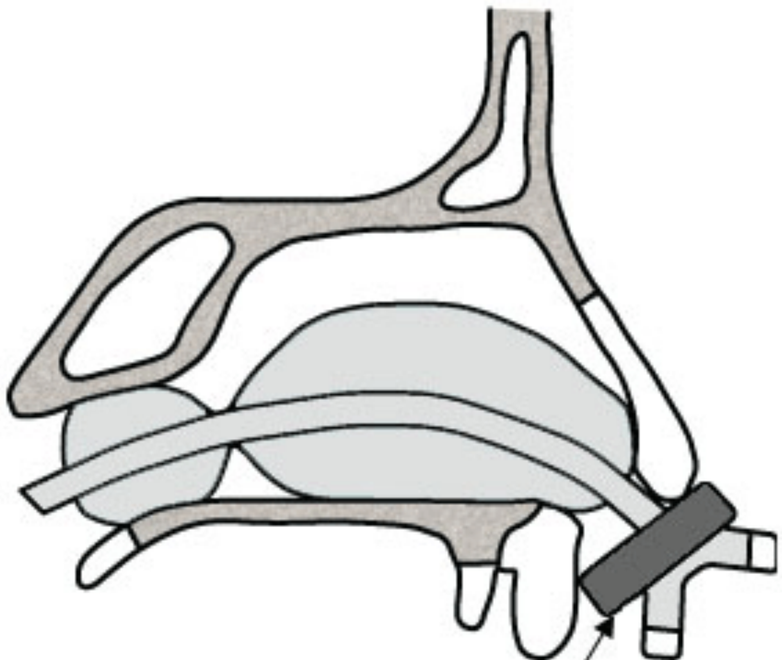
Posterior Nasal Packing

- The patient should be hospitalized
- Indications:
  - -Failure of anterior packing
  - -High suspicion of posterior bleeding

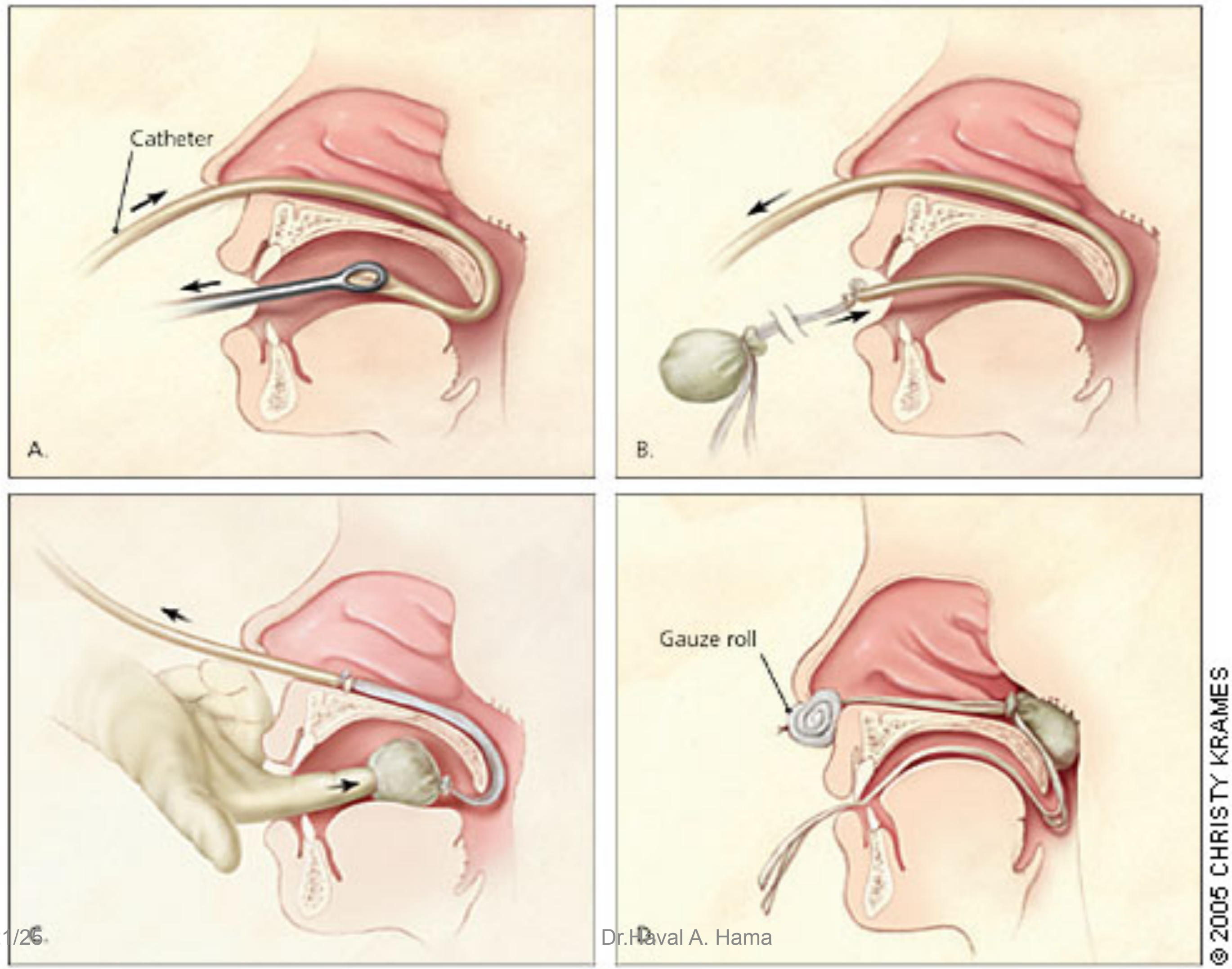
- Foley catheter



- Double-balloon catheter



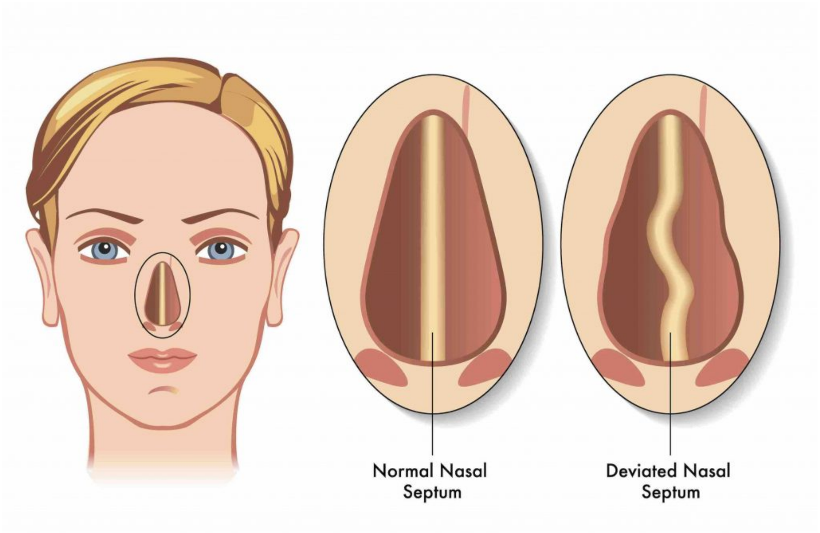
- Gauze method



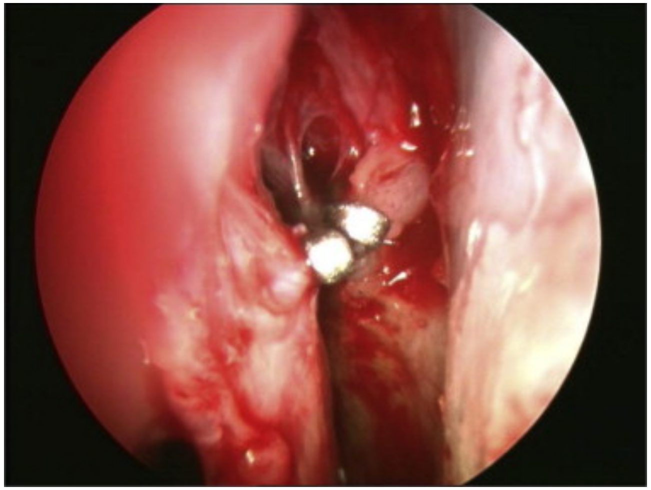
Complication of nasal packing

- Pain and discomfort
- Vasovagal attack
- hypoxia,
- Blockage of nasolacrimal duct and epiphora
- Eustachian tub dysfunction
- Toxic shock syndrome
- Septal necrosis and perforation
- Bleeding and injury during instrumentation
- Dislodgment into pharynx and airway obstruction

Surgical Intervention



- Indications:
  - Bleeding continues despite adequate packing and resuscitation



  - Nasal abnormality (septal deviation)
  - Arterial ligation(first artery is SP artery)
- Angiography and vessel embolization (can not be done for anterior and posterior ethmoid because risk of CVA

Medical Treatment:

- Treatment of underlying medical conditions like coagulopathy, renal impairment…. etc.
- Tranexamic acid locally or systemically up to 1.5g three times a day.
- Correction of underlying bleeding disorder.
- Local vasoconstrictor and antibiotic.

Nasal fracture

the nose is the most frequently fractured facial bone

-Is the most prominence and delicate structure

• (between 40–50%),3rd most common fracture of the bony skeleton

•Force required to create a fracture of the nasal structure is small, possibly as little as 25 pounds of pressure

CAUSES

- Physical fights
- assaults
- domestic violence.
- Contact sports
- Falls (common in children)
- Motor vehicle accidents
- Falls from syncope or impaired balance in the elderly and etc.

PATHOGENESIS

Direction of the force:

•Frontal direction : in fracture of the lower margin of the nasal bones

• High frontal : nasal orbital ethmoid fracture

• Heavier force : severe flattening or splaying of the nasal bones and fracture of the septum

• Lateral forces : depression of the ipsilateral nasal bone, out fracture of the contralateral nasal bone

Children’s noses: mostly greenstick fracure

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Classification

CLASS 1	CLASS 2	CLASS 3
CHEVALLET FRACTURE mostly depressed fractures of nasal bones	JARAWAY FRACTURE These # cause a significant amount of cosmetic deformity.	high velocity trauma. AKA nasoorbital #/naso ethmoidal #..
The # line runs parallel to the dorsum of the nose and naso maxillary suture and joins at a point where the nasal bone becomes thicker	In this group not only the nasal bones are fractured, the underlying fronto nasal process of the maxilla is also fractured. Ethmoidal labyrinth & orbit is intact	These fractures are always associated with Le Fort fracture of the upper face involving the maxilla also Ethmoid being thin & full of air cells forms a low resistence 'crumple zone"
Clinically :present as a depression over the nasal bone area. tenderness and crepitus over the affected nasal bone. Radiological evidence may or may not be present. clinical diagnosis		Clinically : telecanthus is seen & medial canthal ligament may be disrupted from lacrimal crest. Epiphora can also be there due to lacrimal crest injury.

SYMPTOMS

- Bruising of the skin and subcutaneous tissues
- pain
- Swelling
- Deformity
- Difficulty breathing
- Nosebleeds
- Deviation and asymmetry
- Epiphora

PHYSICAL EXAMINATION

- Acute edema may hide deformities; however, a careful search for intranasal injury must take place
- Nose should be externally observed from all angles
- Bleeding can be controlled with topical cotton pledgets soaked in vasoconstrictors – 0.25% phenylephrine – 4% cocaine, which also provides anesthesia.
- Nasal airway evaluation :

(a) Obstruction

(b) Hematoma

(c) Septal deviation

- Other signs: –

(a) Oedema

(b)Skin laceration

(c)Periorbital ecchymosis

(d)cerebrospinal fluid (CSF) rhinorrhoea

(e) Olfactory disorders

- Palpation : Evidence of nasal fracture
- Mobility : of the nasal bones is assessed by grasping the dorsum btw two fingers and firmly rocking the pyramid back & forth
- Crepitations can also be felt

- Assessment of nasal cavity may provide additional information and rule out the following: –

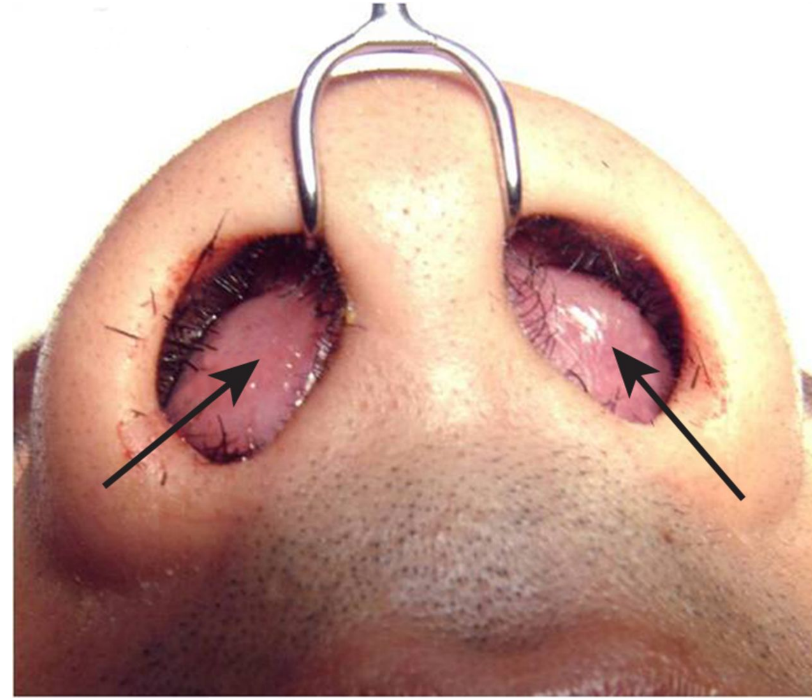
  (a)Mucosal tears

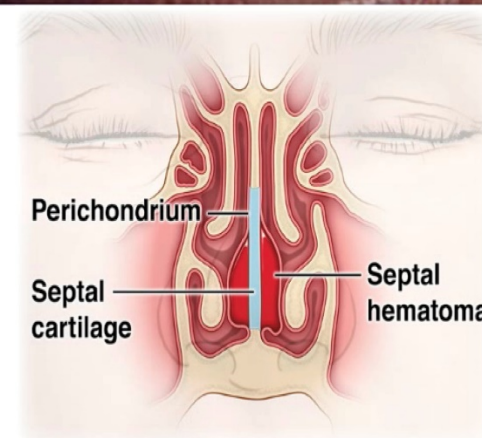
  (b) Lacerations

  (c)Ecchymosis

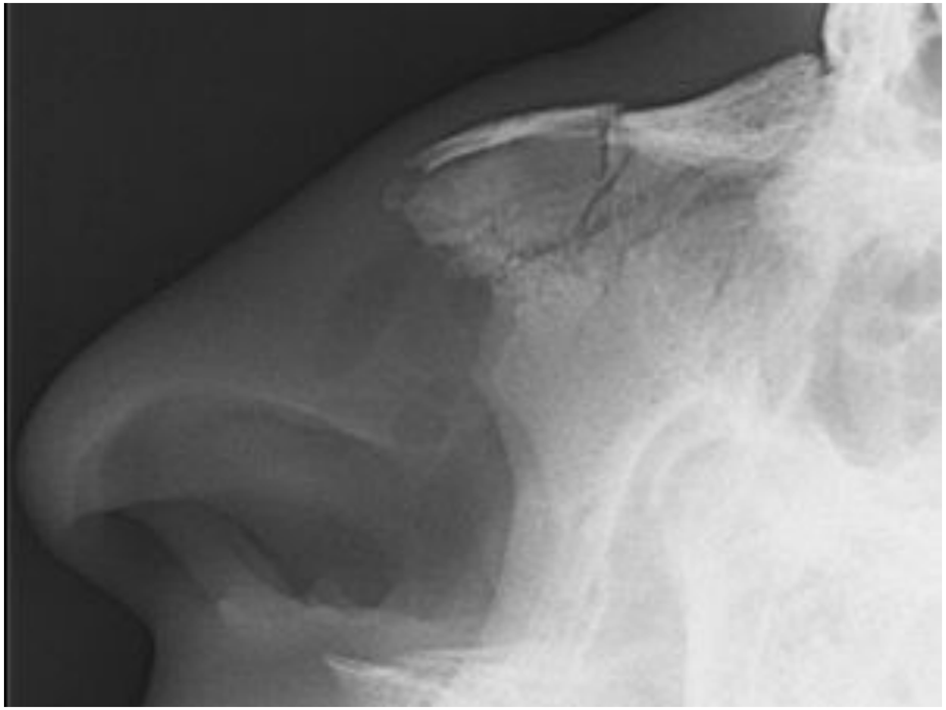
  (d)Hematoma

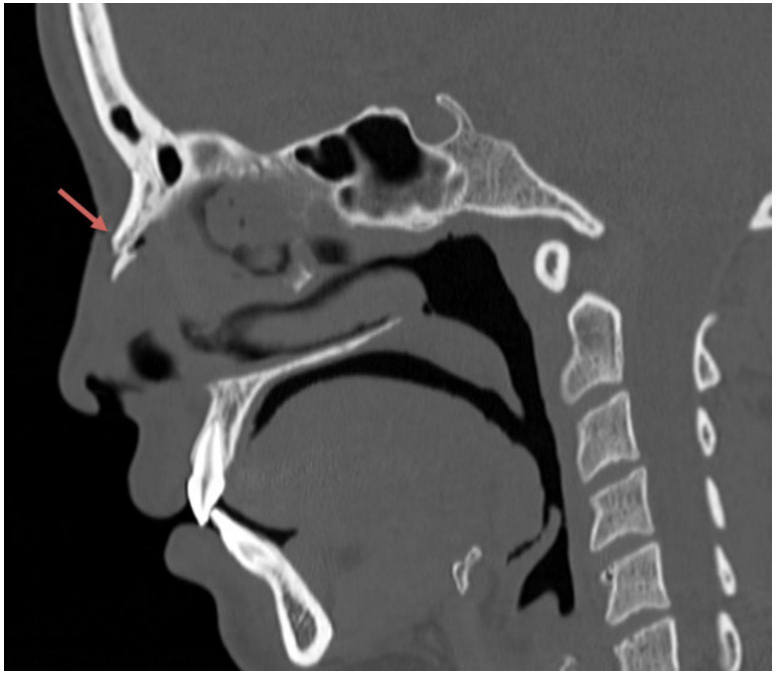
- Push the tip of the nose upward to check for integrity of the septal support system.
- Retained blood clots should be removed





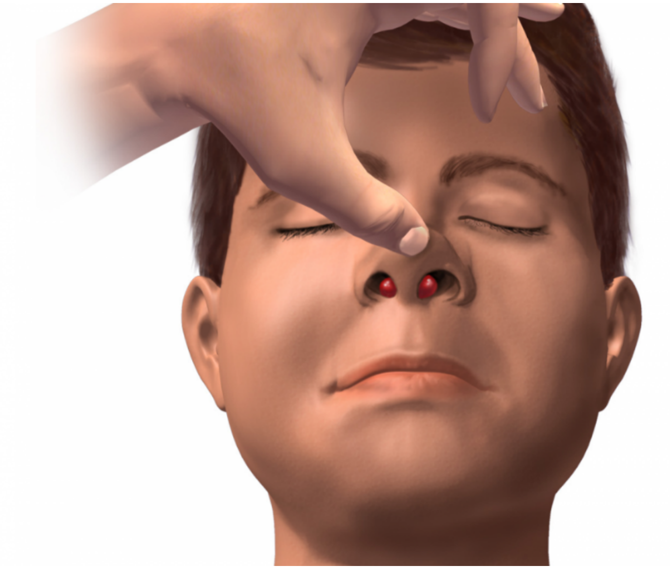
Investigation





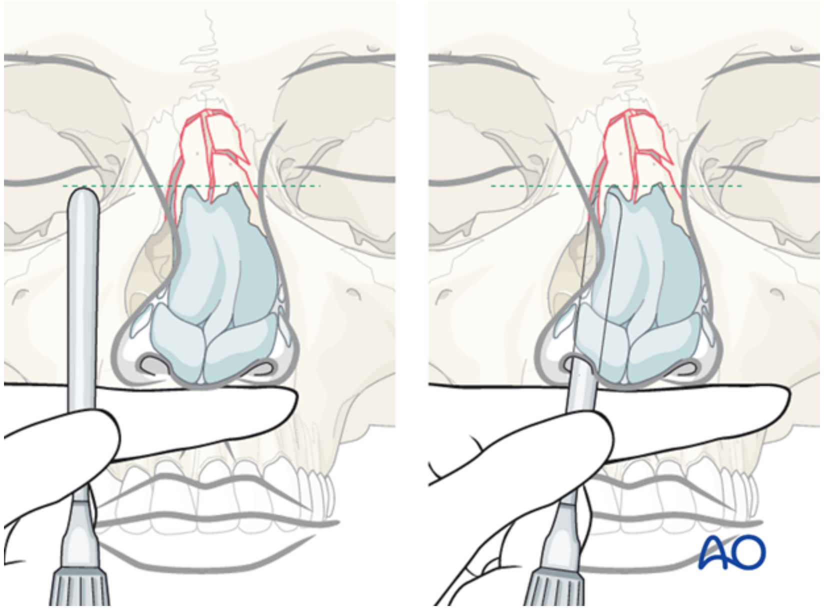
Management

- Advanced Trauma Life Support ,an assessment of the ABCs (airway, breathing, and circulation)
- Nose bleeding & septal hematoma should be assessed.
- Check for other facial fractures, e.g. orbital rim, mandible
- Treatment begins with management of external soft tissue injuries (clean lacerations and carefully repair them)



TIMING OF REDUCTION

- The development of fibrous connective tissue within the fracture line becomes the limiting factor starting at around 10 days to 2 weeks after injury.
- Hence the best time is to start before this phase.
- A short period of delay is recommended during first 2-3 days to allow diminution of oedema so that nasal bone position is best appreciated.



	Closed reduction	Open reduction
	CR involves manipulation of the nasal bones without incision & is generally the preferred choice	OR may include a range of techniques including septoplasty , osteotomy or septorhinoplasty.
Indication:	• Unilateral / bilateral # of nasal bones • • • # of nasal septal complex with nasal deviation of less than half of the width of nasal bridge	• Bilateral # with dislocation of nasal dorsum & significant pathological changes • Infracture of nasal dorsum

COMPLICATIONS

Early	Late
• Septal hematoma	1. Untreated hematoma : lead to sub perichondrial fibrosis & partial nasal airway obstruction
2. Edema	2. synechiae
3. Ecchymosis	3. Residual osteitis
4.Csf leak	4. Chronic rhinosinusitis
5. Epistaxis	5. Cosmetic Deformity

Conclusions

- Epistaxis is the most common rhinological emergency condition
- minority of the case require surgery
- Old age with chronic medical condition require hospitalization
- Nasal bone fracture is the most common facial fracture
- Septal hematoma should be excluded in patient with nasal trauma
- Most of fracture nose in childereen is greenstick fracture
- Time for reduction is very important for a good result

Source

- LOGAN TURNER